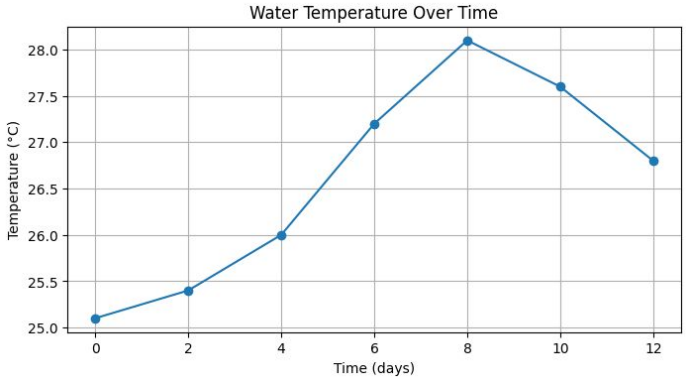
Lab 7.1: Plot Time vs. Temperature

Task: Visualize the following measurements.

```
time = [0, 2, 4, 6, 8, 10, 12]
temperature = [25.1, 25.4, 26.0, 27.2, 28.1, 27.6, 26.8]
```



1. Create a line plot.
2. Set the figure size to (8, 4).
3. Add a meaningful title and axis labels.
4. Include the time (days) & temperature (°C) unit.
5. Add markers and gridlines.
6. Display the figure.

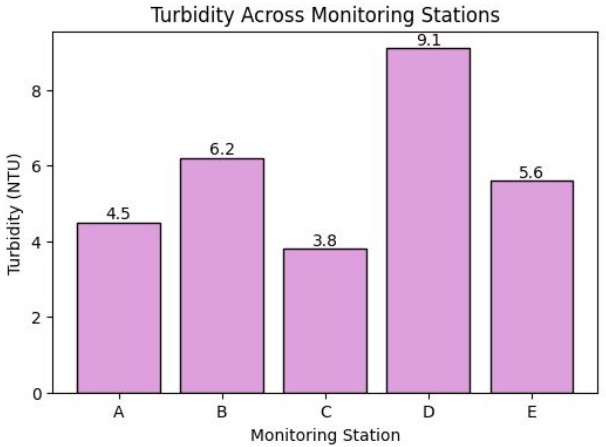


Lab 7.2: Create a Bar Chart

Task: Visualize the following measurements.

```
stations = ["A", "B", "C", "D", "E"]
turbidity = [4.5, 6.2, 3.8, 9.1, 5.6]
```

1. Create a bar chart of turbidity by station.
2. Add a meaningful title and axis labels.
3. Change the bar color.
4. Add numerical labels to the bars.
5. Display the figure.





Lab 7.3: Exploratory Data Analysis (EDA) with Seaborn

Exploratory Data Analysis (EDA) uses summary statistics and visualizations to understand a dataset before further analysis or modeling.

Load the Dataset

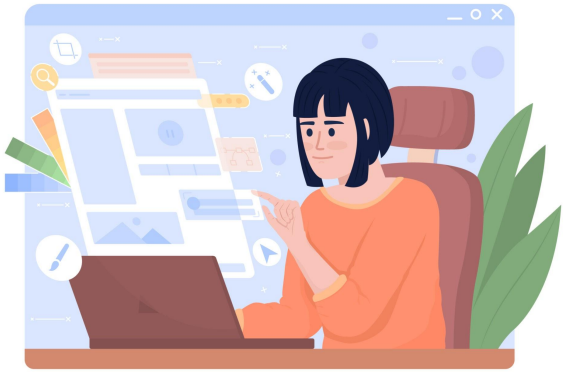
</> Python

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(
   "/content/river_water_quality.csv"
)

print(df.head())
```

	Timestamp	Conductivity	NO3	Temp	Turbidity	Dayofweek	Month
0	2017-05-11T14:00:00	13502.10	NaN	NaN	12.79	3	5
1	2017-05-11T15:00:00	10304.00	NaN	NaN	11.93	3	5
2	2017-05-11T16:00:00	5588.08	NaN	NaN	10.34	3	5
3	2017-05-11T17:00:00	13937.00	NaN	NaN	24.02	3	5
4	2017-05-11T18:00:00	44761.40	NaN	NaN	18.46	3	5



Understand the Dataset Structure

</> Python

```
print(df.shape)
print(df.columns)
print(df.info())
```

```
(25354, 7)
Index(['Timestamp', 'Conductivity', 'NO3', 'Temp', 'Turbidity', 'Dayofweek',
       'Month'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25354 entries, 0 to 25353
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Timestamp     25354 non-null  object
 1   Conductivity  16037 non-null  float64
 2   NO3           6651 non-null   float64
 3   Temp          9816 non-null   float64
 4   Turbidity     16037 non-null  float64
 5   Dayofweek     25354 non-null  int64
 6   Month         25354 non-null  int64
dtypes: float64(4), int64(2), object(1)
memory usage: 1.4+ MB
None
```

</> Python

```
missing_count = df.isnull().sum()

print(missing_count)

missing_percent = (
    df.isnull().mean() * 100
).round(2)

print(missing_percent)
```

```
Timestamp          0
Conductivity    9317
NO3            18703
Temp           15538
Turbidity       9317
Dayofweek          0
Month              0
dtype: int64
Timestamp       0.00
Conductivity   36.75
NO3            73.77
Temp           61.28
Turbidity      36.75
Dayofweek       0.00
Month           0.00
dtype: float64
```



Lab 7.3: EDA with Seaborn

Task: Visualize Missing Data

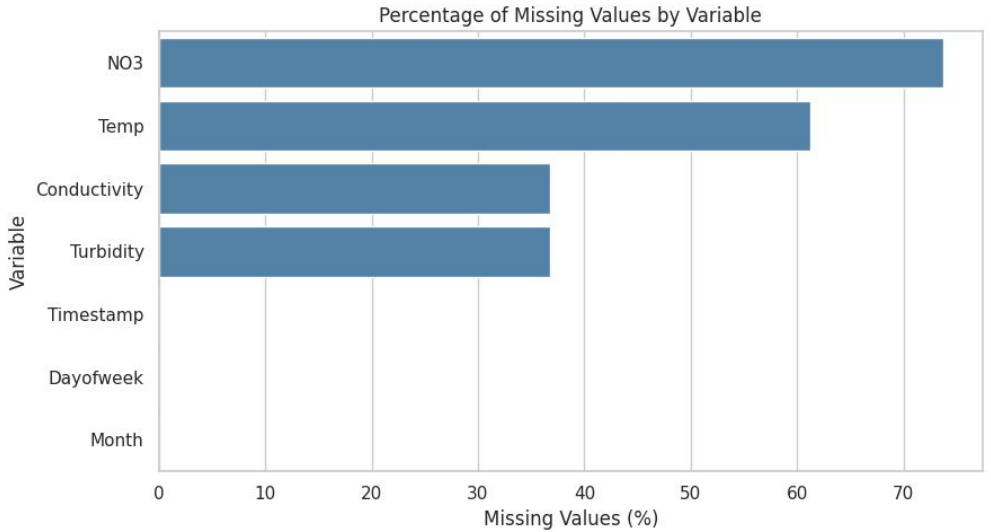
```
missing_percent = (
    df.isnull().mean() * 100
).sort_values(ascending=False)

plt.figure(figsize=(9, 5))

sns.barplot(
    x=missing_percent.values,
    y=missing_percent.index,
    color="steelblue"
)

plt.title("Percentage of Missing Values by Variable")
plt.xlabel("Missing Values (%)")
plt.ylabel("Variable")

plt.tight_layout()
plt.show()
```



Convert the Timestamp to Datetime

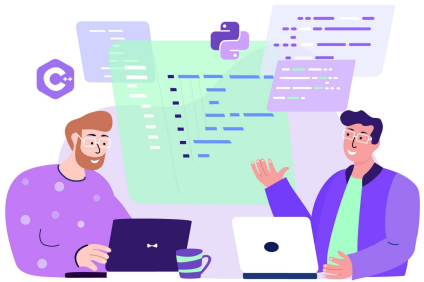
To visualize changes over time, the timestamp should be interpreted as a datetime variable.

</> Python

```
df["Timestamp"] = pd.to_datetime(
    df["Timestamp"],
    errors="coerce"
)

print(df["Timestamp"].dtype)
print(df["Timestamp"].head())
```

```
datetime64[ns]
0   2017-05-11 14:00:00
1   2017-05-11 15:00:00
2   2017-05-11 16:00:00
3   2017-05-11 17:00:00
4   2017-05-11 18:00:00
Name: Timestamp, dtype: datetime64[ns]
```



Lab 7.3: EDA with Seaborn

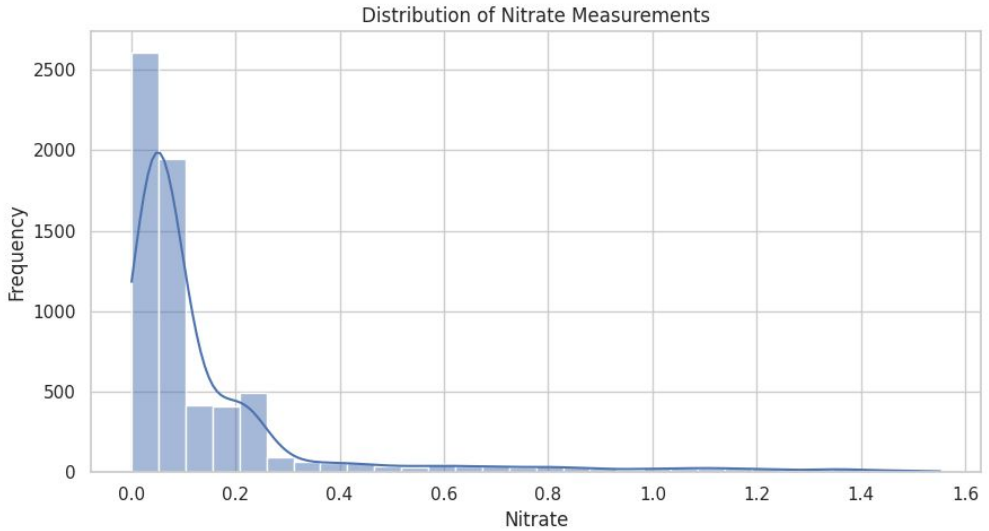
Task: Visualize the Distribution of Nitrate

```
plt.figure(figsize=(9, 5))

sns.histplot(
    data=df,
    x="NO3",
    bins=30,
    kde=True
)

plt.title("Distribution of Nitrate Measurements")
plt.xlabel("Nitrate")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Lab 7.3: EDA with Seaborn

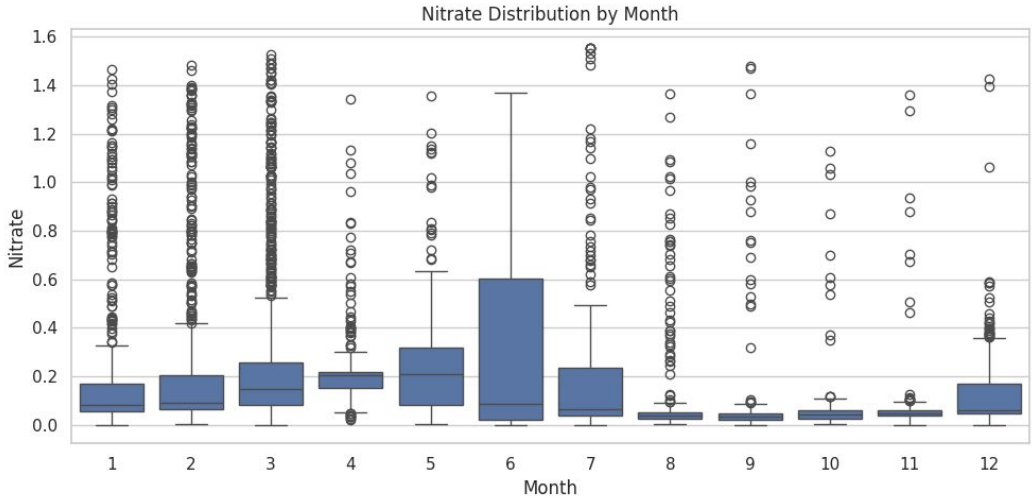
Task: Compare Nitrate Concentrations by Month

```
plt.figure(figsize=(10, 5))

sns.boxplot(
    data=df,
    x="Month",
    y="NO3"
)

plt.title("Nitrate Distribution by Month")
plt.xlabel("Month")
plt.ylabel("Nitrate")

plt.tight_layout()
plt.show()
```



Lab 7.3: EDA with Seaborn

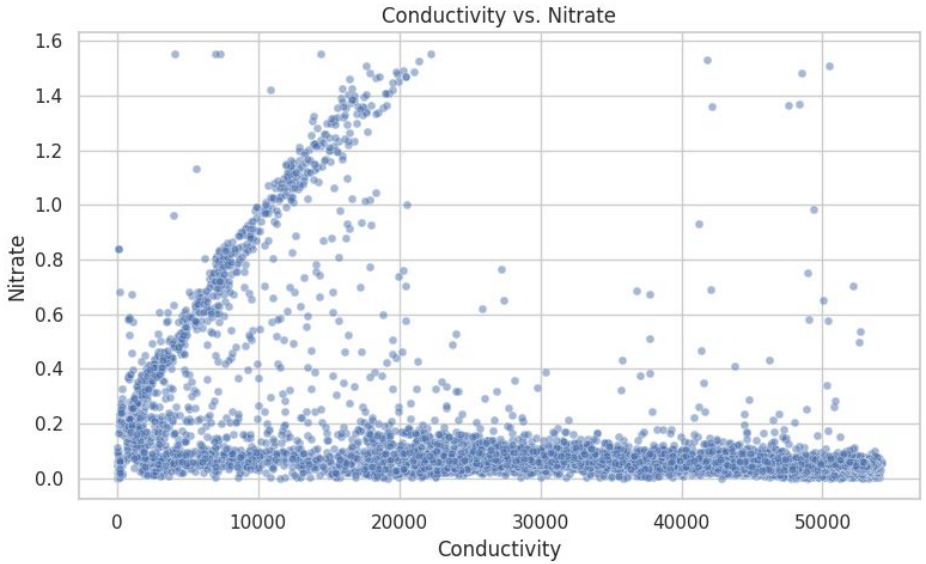
Task: Explore Conductivity and Nitrate

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Conductivity",
    y="NO3",
    alpha=0.5,
    s=25
)

plt.title("Conductivity vs. Nitrate")
plt.xlabel("Conductivity")
plt.ylabel("Nitrate")

plt.tight_layout()
plt.show()
```



Lab 7.3: EDA with Seaborn

Task: Visualize Nitrate Over Time

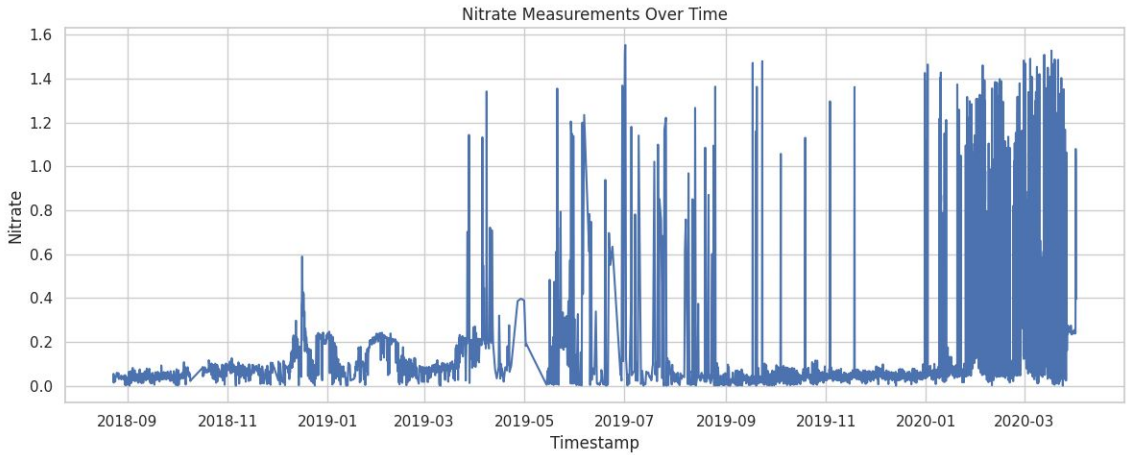
```
time_df = df.dropna(
    subset=["Timestamp", "NO3"]
).sort_values("Timestamp")

plt.figure(figsize=(12, 5))

sns.lineplot(
    data=time_df,
    x="Timestamp",
    y="NO3",
    errorbar=None
)

plt.title("Nitrate Measurements Over Time")
plt.xlabel("Timestamp")
plt.ylabel("Nitrate")

plt.tight_layout()
plt.show()
```



Lab 7.3: EDA with Seaborn

Task: Examine Monthly Patterns

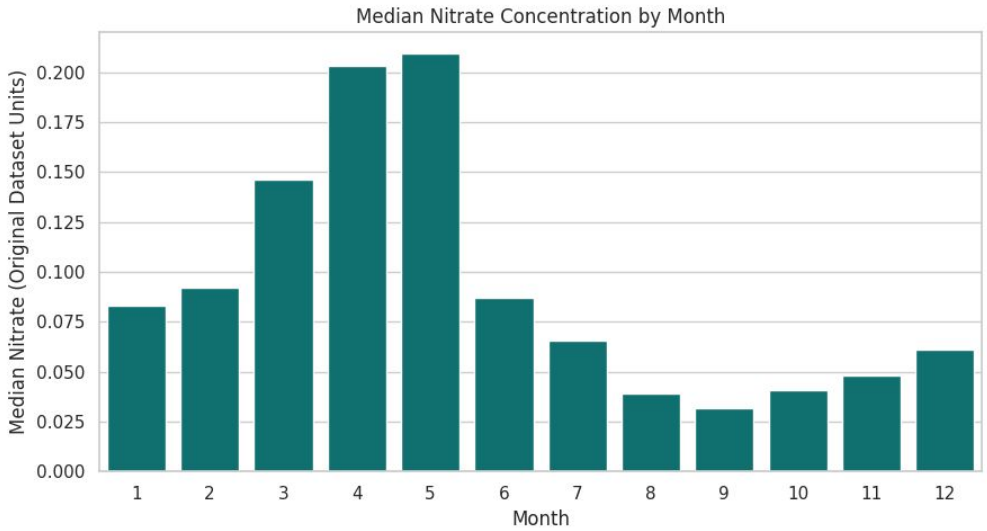
```
monthly_nitrate = (
    df.groupby("Month", dropna=False)["NO3"]
      .median()
      .dropna()
)

plt.figure(figsize=(9, 5))

sns.barplot(
    x=monthly_nitrate.index,
    y=monthly_nitrate.values,
    color="teal"
)

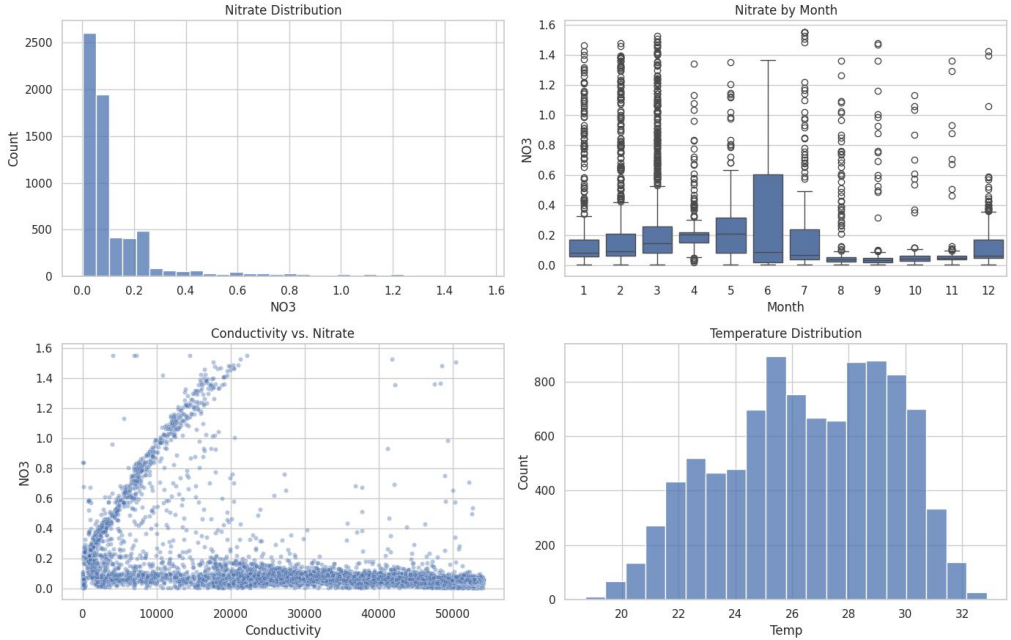
plt.title("Median Nitrate Concentration by Month")
plt.xlabel("Month")
plt.ylabel("Median Nitrate (Original Dataset Units)")

plt.tight_layout()
plt.show()
```



Lab 7.3: EDA with Seaborn

Task: Create a Multi-Panel Plot




```
fig, axes = plt.subplots(2, 2, figsize=(14, 9))

sns.histplot(
    data=df,
    x="NO3",
    bins=30,
    ax=axes[0, 0]
)
axes[0, 0].set_title("Nitrate Distribution")

sns.boxplot(
    data=df,
    x="Month",
    y="NO3",
    ax=axes[0, 1]
)
axes[0, 1].set_title("Nitrate by Month")

sns.scatterplot(
    data=df,
    x="Conductivity",
    y="NO3",
    alpha=0.4,
    s=20,
    ax=axes[1, 0]
)
axes[1, 0].set_title("Conductivity vs. Nitrate")

sns.histplot(
    data=df,
    x="Temp",
    bins=20,
    ax=axes[1, 1]
)
axes[1, 1].set_title("Temperature Distribution")

plt.tight_layout()
plt.show()
```

Lab 7.3: EDA with Seaborn

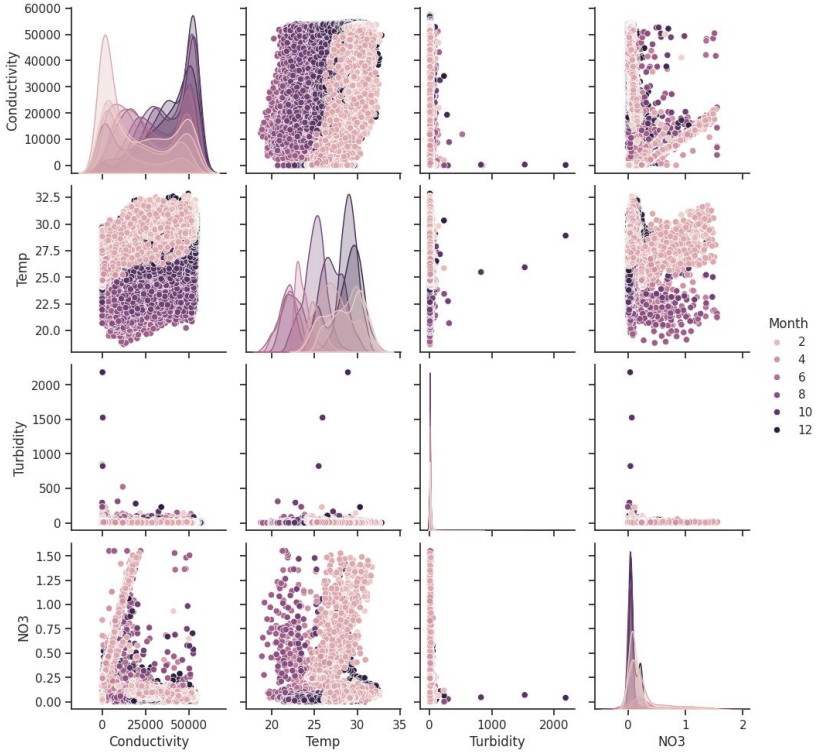
Task: Create a Multi-Panel Plot (Pair Plot)

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="ticks")

data = df[["Conductivity", "Temp", "Turbidity", "NO3", "Month"]]

sns.pairplot(data, hue="Month")
plt.show()
```



Extra: Interactive Chart

</> Python

```
%pip install plotly
```

</> Python

```
import plotly.express as px
import plotly.graph_objects as go
import pandas as pd
```



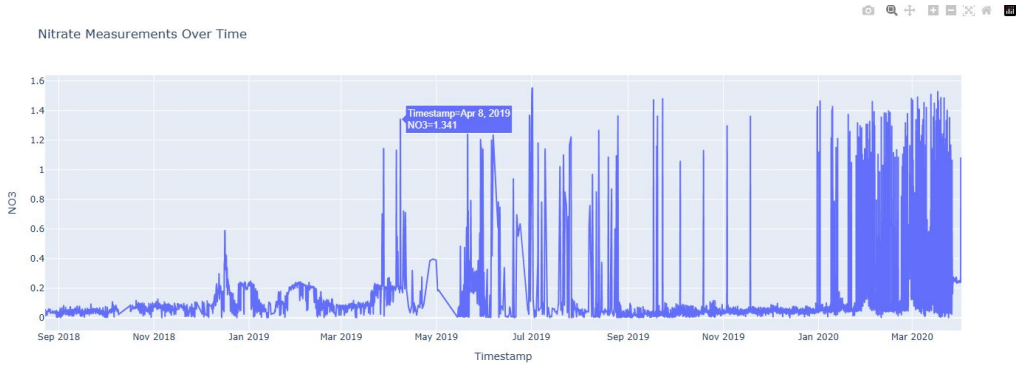
Extra: Interactive Chart

Task: Interactive Time-series Chart

```
time_df = (
    df.dropna(subset=["Timestamp", "NO3"])
      .sort_values("Timestamp")
)

fig = px.line(
    time_df,
    x="Timestamp",
    y="NO3",
    title="Nitrate Measurements Over Time"
)

fig.show()
```



Extra: Interactive Chart

Task: Interactive Scatter Matrix

```
df["Date"] = df["Timestamp"].dt.date
df["Hour"] = df["Timestamp"].dt.hour

variables = ["Conductivity", "NO3", "Temp", "Turbidity"]

for col in variables:
    df[col] = pd.to_numeric(df[col], errors="coerce")
```




```
plot_df = df[variables].dropna()

fig = px.scatter_matrix(
    plot_df,
    dimensions=variables,
    title="Pairwise Relationships Between Water Quality Variables",
    opacity=0.5
)

fig.update_layout(
    width=1000,
    height=850
)

fig.show()
```